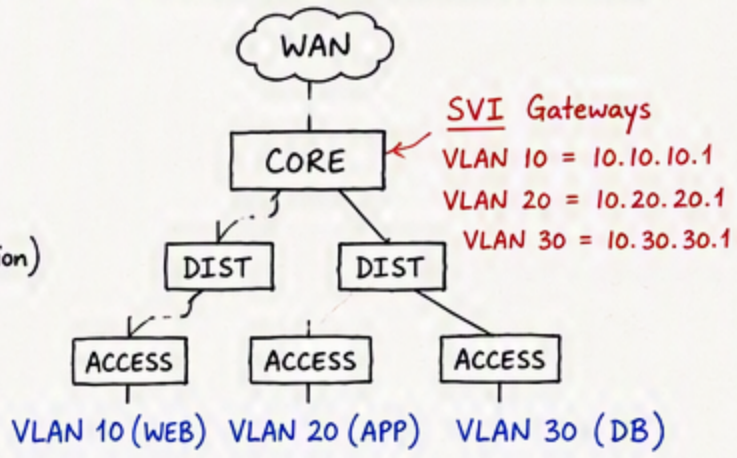
ACI NOTES – PART 4

LEGACY TO ACI MIGRATION

1. WHY MIGRATE TO ACI ?

- ✓ Centralized Policy
- ✓ Application Centric
- ✓ Automation
- ✓ Visibility
- ✓ Better Security (Microsegmentation)
- ✓ Scalability
- ✓ Operational Simplicity

2. LEGACY NETWORK EXAMPLE



3. MIGRATION PRINCIPLES

- ☆ Move in Waves (Not Big Bang)
- ☆ Start Network-Centric
- ☆ Understand Traffic First
- ☆ Keep it Simple Initially
- ☆ Minimize Application Impact
- ☆ Validate Each Step
- ☆ Have Rollback Plan

4. MIGRATION MODELS

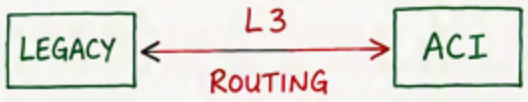
A. L2 STRETCH (COMMON FOR MIGRATION)



PROS:
- Same VLAN / Subnet
- No IP changes
- Low Application Impact

CONS:
- Large Broadcast Domain
- Bigger Failure Domain
- STP Considerations
- Harder Troubleshooting

B. L3 MIGRATION (ARCHITECT GOAL)



PROS:
- Smaller Failure Domain
- Easier Troubleshooting
- Better Scalability
- Cleaner Design

CONS:
- Requires Routing Changes
- More Planning

5. MIGRATION PHASES (TYPICAL)

1. Build ACI Fabric (Spine, Leaf, APIC)
2. Connect Legacy Core to ACI (L2 or L3)
3. Create BDs / EPGs (Network-Centric)
4. Stretch VLANs (if using L2 Model)
5. Migrate First Hop Gateway to ACI
6. Move Servers / Workloads in Waves
7. Traffic Discovery & Monitoring
8. Create Contracts (Application-Centric)
9. Remove Preferred Group / Permit-All
10. Decommission Legacy

6. FIRST HOP GATEWAY MIGRATION (L2 MODEL)

BEFORE:



AFTER:

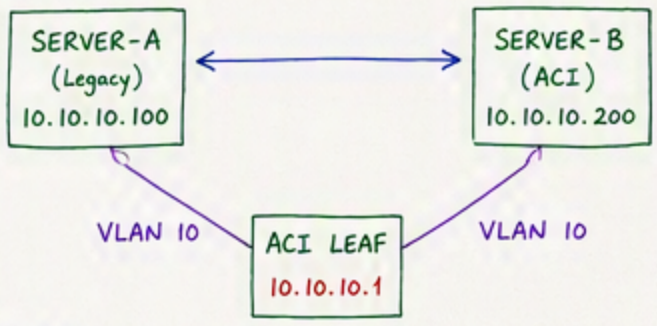


STEPS:

1. Prepare BD in ACI
   BD-WEB = 10.10.10.1/24
   Unicast Routing = Disabled
2. Shutdown Gateway on Legacy Core
3. Enable Gateway in ACI BD
   Unicast Routing = Enabled
4. Hosts ARP for 10.10.10.1 (ACI Now Replies)

NOTES:
- Never have same gateway active on Legacy and ACI.
- Ensure VLAN is allowed on Migration Link.

7. COMMUNICATION DURING MIGRATION (L2 STRETCH)



FLOW:
1. Server-A ARPs for Server-B IP (10.10.10.200)
2. ARP Request travels over stretched VLAN
3. ACI Leaf already learned Server-B (EPm)
4. ACI replies via Proxy ARP with MAC-B
5. Traffic forwarded normally to Server-B

8. POLICY DURING MIGRATION

Phase 1 (Initial):
- Use Preferred Group OR Permit-All Contracts
- Goal = Stability, No Outage

Phase 2 (Discovery):
- Use Tools (Flow, Logs, Splunk, Firewall Logs, SPAN)
- Build Traffic Matrix (Who talks to whom, What ports)

Phase 3 (Application Centric):
- Create Contracts based on real traffic

Phase 4 (Hardening):
- Validate Contract Hit Counters
- Remove Preferred Group / Permit-All

Rollback Plan:
- Re-enable Preferred Group if issue occurs

9. TRAFFIC DISCOVERY METHODS

- NetFlow / IPFIX
- Firewall Logs
- Splunk / Log Analytics
- ACI Contract Hit Counters
- Application Monitoring Tools
- SPAN / Packet Capture
- Application Team Validation

GOAL:
Build accurate port/protocol list before creating contracts.

10. COMMON MIGRATION ISSUES

1. Duplicate Gateway (Old + New Active)
2. VLAN not allowed on Migration Link
3. Contract Missing (Policy Drop)
4. Route missing on Leaf (RCP / Redistribution)
5. External EPG mismatch
6. Asymmetric Routing
7. Service Graph misconfiguration

KEY COMMANDS (EXAMPLES)

- show endpoint ip <ip>
- show endpoint mac <mac>
- show ip route vrf <vrf>
- show zoning-rule
- show l3out
- show bgp l2vpn evpn summary
- show bgp l2vpn evpn route-type 2
- show bgp l2vpn evpn route-type 5
- show service-graph
- show contract | include <contract>

MIGRATION GOLDEN RULES

1. Move Applications, not VLANs.
2. Start Simple, then Harden.
3. Validate Routing + Policy.
4. Always have a Rollback Plan.

MIGRATION SUCCESS =

No Outage +
Happy Application Team +
Better Security +
Stable & Scalable ACI Fabric

REMEMBER:

Routing gets you there,
Policy decides if you can talk.

☺